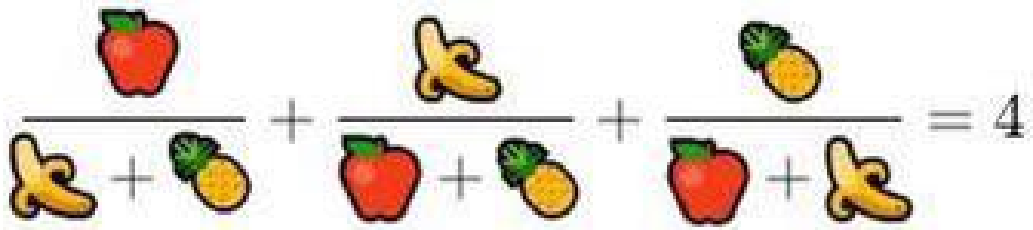
Elliptic Fruit: An Intro to Elliptic Curves

Yuval Amit and Noah Kupinsky

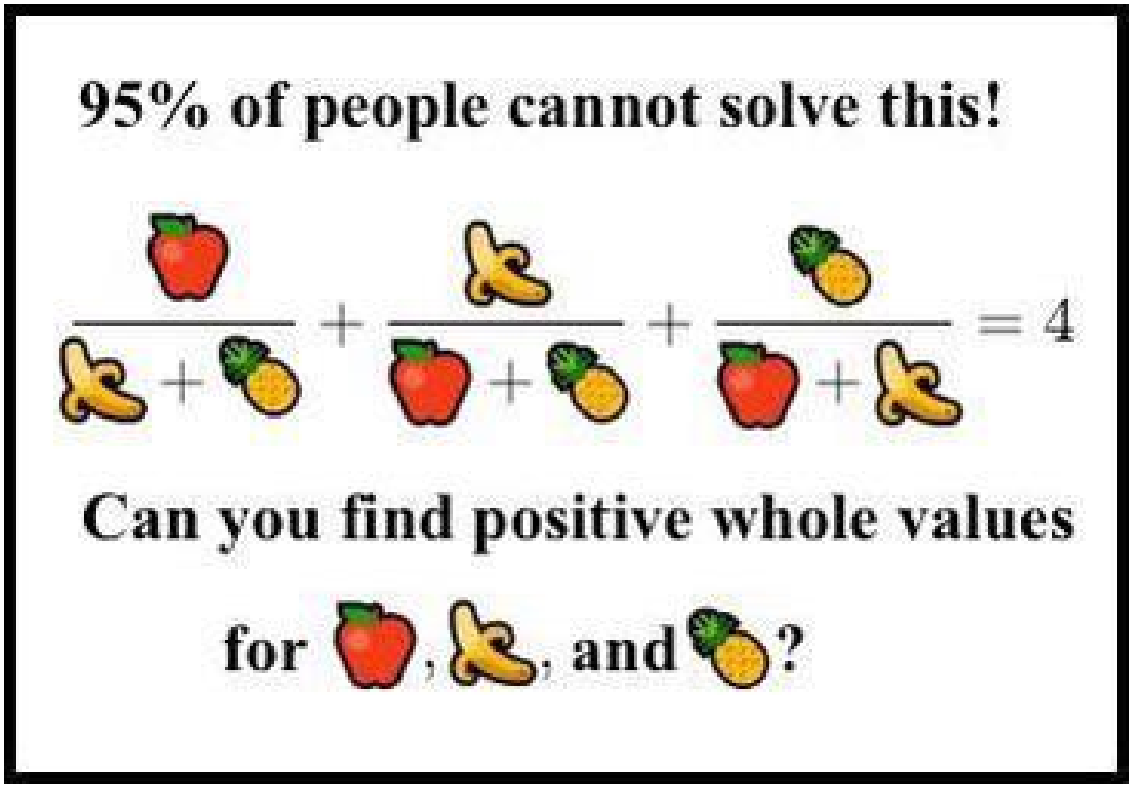
A Silly Problem

95% of people cannot solve this!



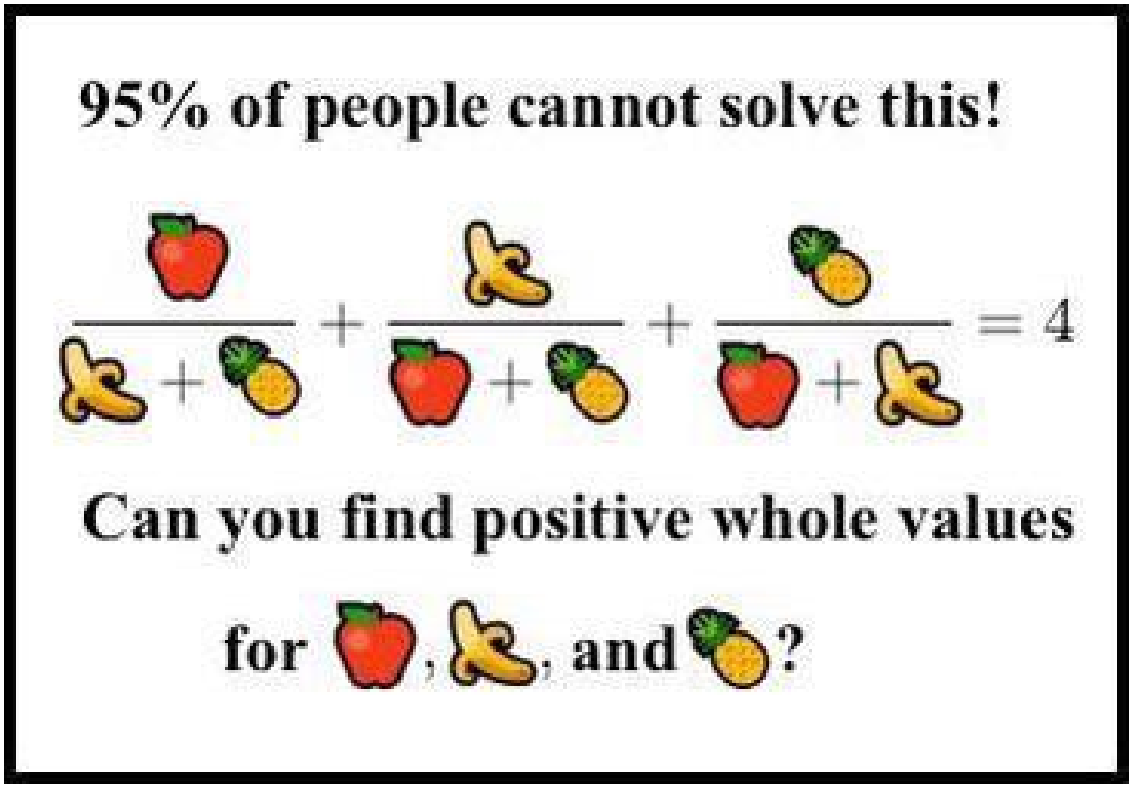
Can you find positive whole values for  **,**  **, and**  **?**

A Silly Problem



Actually, about 99.9999% of people can't solve this.

A Silly Problem



Actually, about 99.9999% of people can't solve this.

Let's solve it.

How to Solve

Problem

Find $X, Y, Z \in \mathbb{Z}^+$ such that

$$\frac{X}{Y+Z} + \frac{Y}{X+Z} + \frac{Z}{X+Y} = 4$$

How to Solve

Problem

Find $X, Y, Z \in \mathbb{Z}^+$ such that

$$\frac{X}{Y+Z} + \frac{Y}{X+Z} + \frac{Z}{X+Y} = 4$$

Observation 1

If (X, Y, Z) is a solution, then so is (kX, kY, kZ) for any $k \in \mathbb{R}$. We may therefore set $z = 1, x = \frac{X}{Z}, y = \frac{Y}{Z}$ and search for positive rational solutions $(x, y, 1)$ to the simpler equation

$$\frac{x}{y+1} + \frac{y}{x+1} + \frac{1}{x+y} = 4 \iff (x+1)(y+1) + x(x+1)(x+y)$$

$$+y(y+1)(x+y) - 4(x+y)(x+1)(y+1) = 0$$

Cubic Curves

Observation 2

The equation we obtained is not much nicer, but it transforms the problem into something more well understood: finding rational points on **cubic curves**.

Cubic Curves

Observation 2

The equation we obtained is not much nicer, but it transforms the problem into something more well understood: finding rational points on **cubic curves**.

Let $f(x, y)$ be a cubic polynomial in x and y with rational coefficients, and a cubic curve is the solution set in $\mathbb{R}^2$ of $f(x, y) = 0$.

Cubic Curves

Observation 2

The equation we obtained is not much nicer, but it transforms the problem into something more well understood: finding rational points on **cubic curves**.

Let $f(x, y)$ be a cubic polynomial in x and y with rational coefficients, and a cubic curve is the solution set in $\mathbb{R}^2$ of $f(x, y) = 0$.

Let $(a_1, b_1), (a_2, b_2)$ be rational points on the curve. Consider the line which passes through them, say $y = mx + b$. Let $P(x) = f(x, mx + b)$, and we know $P(a_1) = P(a_2) = 0$.

Cubic Curves

Observation 2

The equation we obtained is not much nicer, but it transforms the problem into something more well understood: finding rational points on **cubic curves**.

Let $f(x, y)$ be a cubic polynomial in x and y with rational coefficients, and a cubic curve is the solution set in $\mathbb{R}^2$ of $f(x, y) = 0$.

Let $(a_1, b_1), (a_2, b_2)$ be rational points on the curve. Consider the line which passes through them, say $y = mx + b$. Let $P(x) = f(x, mx + b)$, and we know $P(a_1) = P(a_2) = 0$.

The line passing through them the hits curve a third time at another rational point!

More solutions!!!

We want to find more rational points on the curve. We now have three points $(a_1, b_1), (a_2, b_2), (a_3, b_3)$, but they all lie on the same line.

More solutions!!!

We want to find more rational points on the curve. We now have three points $(a_1, b_1), (a_2, b_2), (a_3, b_3)$, but they all lie on the same line.

Observation 3

Now recall our particular cubic curve
$(x+1)(y+1)+x(x+1)(x+y)+y(y+1)(x+y)-4(x+y)(x+1)(y+1)$
It's symmetric! So, if (a_3, b_3) is a solution, so is (b_3, a_3). This gives us a new rational point on a different line, and we can look at more lines to get more rational points.

More solutions!!!

We want to find more rational points on the curve. We now have three points $(a_1, b_1), (a_2, b_2), (a_3, b_3)$, but they all lie on the same line.

Observation 3

Now recall our particular cubic curve
$(x+1)(y+1)+x(x+1)(x+y)+y(y+1)(x+y)-4(x+y)(x+1)(y+1)$
It's symmetric! So, if (a_3, b_3) is a solution, so is (b_3, a_3). This gives us a new rational point on a different line, and we can look at more lines to get more rational points.

Desmos Demonstration

The Solution

Here it is!

$X = 154476802108746166441951315019919837485664325669565431700$

(81 digits)

$Y = 368751317941299998271978115652254748254929799689719709960$

(80 digits)

$Z = 437361267792869725786125260237139015281653755816161361860$

(79 digits)